UniFraud-GH: Semantic Concept Alignment and Domain-Adaptive Learning for Cross-Sector Fraud Detection in Ghana

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***Abstract*—Financial fraud in Ghana spans payment cards, mobile money and national health insurance, yet labelled data exist in useful quantity in at most one sector at a time. We study *heterogeneous cross-sector transfer* for fraud detection: a classifier trained where labels are plentiful is adapted to a sector where they are scarce, even though the sectors share neither features nor class priors. We contribute (i) a *semantic concept alignment* that maps each sector's schema into a shared three-dimensional "fraud-concept" space (amount anomaly, counterparty risk, structural distance) through a z-score–average operator whose scale-invariance and sign-orientation properties we prove; (ii) a unified formulation of three adaptation strategies—domain-adversarial training (DANN), class-conditional subdomain MMD alignment, and an attention-gated network with source pre-training and target fine-tuning (HEN)—together with an analysis, based on a Jensen–Shannon lower bound on joint error, of why invariant-representation methods are expected to fail under the extreme label shift between sectors (0.52% vs. 54.8% fraud); (iii) a leakage-free, label-budget-aware evaluation protocol with calibrated thresholds, resampling-corrected significance tests and Holm correction; and (iv) a formal audit of rule-generated labels, showing that benchmark scores on rule-labelled claims measure rule reproduction rather than fraud detection unless the rule inputs are excluded. The mathematical development is complete; empirical claims are deferred to a corrected re-run.**

***Index Terms*—fraud detection, domain adaptation, transfer learning, class imbalance, maximum mean discrepancy, mobile money, health insurance, explainable AI.**

I. Introduction

FRAUD in Ghana's financial ecosystem is not confined to one channel. Card-not-present attacks, mobile-money social engineering and insurance-claim padding are carried out by overlapping populations of actors using recurring behavioural tactics: abnormal amounts, exploitation of weakly-vetted counterparties, and structural irregularities in time, geography or billing. Labelled examples are nevertheless siloed and uneven: card fraud is abundant but extremely rare per transaction, mobile-money simulators produce balanced traffic, and insurance claims are few and labelled mainly by tariff rules.

This situation calls for transfer across heterogeneous domains with different feature spaces $\mathcal{X}_k$ *and* different label priors. Two facts make it hard. First, conventional domain adaptation presupposes a common input space; second, even with a common representation, invariant-representation learning provably cannot help—and can hurt—when $p_S(y) \neq p_T(y)$ [1]. We therefore build the problem from the ground up and keep every claim tied to a statement that can be proved or tested.

Contributions:

1) **Semantic concept alignment** (Section V-B): a schema-to-concept operator $\phi_k : \mathbb{R}^{d_k} \to \mathbb{R}^3$ with proofs of affine invariance and orientation consistency (Prop. 2).
2) **Unified adaptation formulation** (Section V-E): DANN, class-conditional subdomain MMD and HEN expressed as instances of one objective, with the saddle-point characterisation of the gradient-reversal layer (Prop. 5) and a label-shift lower bound (Prop. 6) evaluated on the actual priors of our three sectors.
3) **Prior-aware decision theory** (Section V-C): exact relations between weighted loss, resampling and the Bayes-optimal cost-sensitive threshold (Props. 3 and 4).
4) **Leakage-free evaluation** (Section VI) including a formal audit of rule-generated labels (Section IV, Prop. 1).

II. Related Work

Fraud detection. Supervised gradient-boosted trees remain the strongest baseline on tabular fraud data [2], [3], with unsupervised detectors such as isolation forests [4] and autoencoders used when labels are unavailable. Realistic evaluation must respect temporal drift and label delay [5]. **Domain adaptation.** Theory goes back to the $\mathcal{H}\Delta\mathcal{H}$-divergence bound [6]; practical methods align features adversarially [7], [8], by kernel moment matching [9], [10], by second-order statistics [11], or class-conditionally [12]. Surveys appear in [13]. Learning invariant representations under label shift is limited by the bound of [1]. **Evaluation hygiene.** Target leakage [14], resampling-induced optimism, and unreliable significance tests on cross-validation folds [15] are common causes of irreproducible results; model selection for unsupervised adaptation must not use target labels [16].

III. Problem Formulation

Let $k \in \mathcal{K} = \{\mathrm{CC}, \mathrm{MM}, \mathrm{HI}\}$ index sectors (credit card, mobile money, health insurance). Sector k has a feature space $\mathcal{X}_k \subseteq \mathbb{R}^{d_k}$, label space $\mathcal{Y} = \{0, 1\}$ (1 = fraud), and joint distribution $\mathcal{D}_k$ over $\mathcal{X}_k \times \mathcal{Y}$. A labelled sample $\mathcal{S}_k = \{(x_i, y_i)\}_{i=1}^{n_k} \sim \mathcal{D}_k^{n_k}$ is available, with prior $\pi_k = \mathbb{P}_{\mathcal{D}_k}(Y = 1)$.

Manuscript draft November 3, 2026. Numerical results in Section VII are placeholders (**TBD**) pending re-execution of the corrected, leakage-free protocol of Section VI.

TABLE I
DATASET STATISTICS (FROM THE CORRECTED PIPELINE'S INVENTORY; VERIFIED AGAINST NOTEBOOK OUTPUTS).

Sector	n_k	π_k	Label provenance
Credit card (Sparkov)	1 852 394	0.521%	simulator ground truth
Mobile money (MoMTSim)	550 358	54.84%	simulator ground truth
Health ins. (NHIA)	2 655	41.62%	*rule-generated*

Definition 1 (Heterogeneous transfer problem)**.** *Given a source sector s and a target sector $t \neq s$ with $\mathcal{X}_s \neq \mathcal{X}_t$ and $\pi_s \neq \pi_t$, together with a* label budget *$b \in \{0\} \cup \mathbb{N}$ of target labels ($b = 0$: unsupervised adaptation), find a predictor $h : \mathcal{X}_t \to [0,1]$ minimising the target risk $R_t(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}_t}\, \ell(h(x), y)$ using $\mathcal{S}_s$, the unlabelled target features, and at most b target labels.*

For a hypothesis class $\mathcal{H}$ over a common space $\mathcal{Z}$, write $\varepsilon_k(h) = \mathbb{P}_{\mathcal{D}_k}(h(z) \neq y)$. The central bound [6] is

$$\varepsilon_t(h) \;\leq\; \varepsilon_s(h) + \tfrac{1}{2}\, d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z) + \lambda^\star, \qquad (1)$$

where $\lambda^\star = \min_{h' \in \mathcal{H}} [\varepsilon_s(h') + \varepsilon_t(h')]$. Eq. (1) is informative only if (a) a common space $\mathcal{Z}$ exists (Section V-B) and (b) $\lambda^\star$ is small, which fails under label shift (Prop. 6).

IV. Data and Label Construction

Sparkov and MoMTSim are synthetic generators [17], [18]; external validity to production data is therefore a stated limitation (Section VIII). The health-insurance claims are real but carry no adjudicated fraud flag; labels are produced by tariff-compliance rules in the spirit of [19].

A. Rule-generated labels

Let a claim have service charge S, medical (drug) charge M, total T and approved tariff A, the latter obtained by joining the claim's G-DRG code to the tariff table (defined only when the join succeeds; match rate 67.1%, 1 781 of 2 655). With $\mathbf{1}[\cdot]$ the indicator and $\mathrm{na}(u) = 0$ for missing u,

$$r_1 = \mathbf{1}[S > A], \quad r_2 = \mathbf{1}[M > \tfrac{1}{2}A], \quad r_3 = \mathbf{1}[T > A],$$
$$r_4 = \mathbf{1}[\,|T - (S+M)| > 1\,], \quad r_5 = \mathbf{1}[\min\{\mathrm{na}(S), \mathrm{na}(M), \mathrm{na}(T)\} \leq 0],$$
$$y = \mathbf{1}\Big[\textstyle\sum_{i=1}^5 r_i > 0\Big]. \qquad (2)$$

Observed trigger counts are $(8, 271, 470, 6, 632)$ for $(r_1, \ldots, r_5)$. Because M is observed for only 2026 of 2655 claims (629 missing), rule r_5 fires on 632 claims: at least $629/632 \approx 99.5\%$ of these triggers are explained by a *missing drug charge*, not by a non-positive bill. Missingness is thus being labelled as fraud, which is a defect of the construction rather than a property of fraud (Section VIII); the corrected pipeline treats r_5 as "data-quality flag" and reports results with and without it.

Proposition 1 (Circularity of rule-generated labels)**.** *Suppose $y = g(x_R)$ for a deterministic g and a feature subset R. If a model class contains a function of x_R that equals g (e.g. a depth-bounded tree for threshold rules), then the Bayes risk over the full feature set is* 0*, and any held-out score on that feature set estimates the* rule-reproduction *ability of the learner rather than fraud detection. Moreover, any feature $f(x_R)$ that is a function of rule inputs inherits this leakage.*

Proof. $h(x) = g(x_R)$ attains zero error under $\mathcal{D}_t$ by construction, so $\min_h R_t(h) = 0$. A held-out estimate of $R_t(\hat{h})$ then measures how closely $\hat{h}$ approximates g. For $f(x_R)$: g is a function of x_R, hence also of $(x_R, f(x_R))$. □

In particular the features S, M, T, A, ratios and deviations $(\cdot - A)/A$, the per-DRG label rate, and quasi-identifiers such as folder and insurance-ID numbers (which allow the same patient to appear in train and test) must be excluded, and splits must be grouped by patient.

V. Methodology

A. Leakage-free preprocessing

Let $\mathcal{I}_{\mathrm{tr}}, \mathcal{I}_{\mathrm{va}}, \mathcal{I}_{\mathrm{te}}$ be a partition of indices (random stratified for HI grouped by patient; *chronological* for CC and MM). Every data-dependent transform T_θ (imputation, capping at the q-th percentile, scaling, categorical encoding, feature selection) is fit on $\mathcal{I}_{\mathrm{tr}}$ only: $\hat{\theta} = \theta(\{x_i\}_{i\in\mathcal{I}_{\mathrm{tr}}})$, then applied unchanged to validation and test. For a categorical value c, the smoothed target encoding is

$$\mathrm{TE}(c) = \frac{n_c \bar{y}_c + m\, \bar{y}}{n_c + m}, \qquad (3)$$

[20], computed from *out-of-fold* statistics within the training partition so that a row's own label never enters its encoding. Resampling (below) is applied inside each training fold only.

B. Semantic concept alignment

Let $F_k = \{1, \ldots, d_k\}$ index sector-k features. A domain expert (or a documented rule table) provides, for each concept $c \in \{\text{amt, cpty, dist}\}$, a disjoint index set $G_{kc} \subseteq F_k$ and an orientation $\sigma_j \in \{-1, +1\}$ for $j \in G_{kc}$ (+1 iff larger values indicate higher risk).

Definition 2 (Concept map)**.** *Let $\mu_j, s_j > 0$ be the training mean and standard deviation of feature j. The concept map $\phi_k : \mathbb{R}^{d_k} \to \mathbb{R}^3$ has components*

$$\phi_k(x)_c = \frac{1}{|G_{kc}|} \sum_{j \in G_{kc}} \sigma_j\, \frac{x_j - \mu_j}{s_j}, \qquad \phi_k(x)_c := 0 \text{ if } G_{kc} = \emptyset. \qquad (4)$$

Proposition 2 (Affine invariance and orientation consistency)**.** *(i) If feature j is replaced by $a_j x_j + b_j$ with $a_j > 0$ (a change of units, e.g. USD→GHS), then $\phi_k(x)_c$ is unchanged. (ii) Each component has training mean* 0*, and, if features in G_{kc} have pairwise correlations ρ_{jl} (after orientation), $\mathrm{Var}(\phi_k(x)_c) = |G_{kc}|^{-2} \sum_{j,l\in G_{kc}} \rho_{jl}$, which lies in $[\bar{\rho}, 1]$ with $\bar{\rho}$ the mean pairwise correlation. (iii) If all oriented features in G_{kc} are monotone non-decreasing in a latent risk ρ, then $\phi_k(x)_c$ is monotone non-decreasing in ρ.*

Proof. (i) With $\mu_j' = a_j\mu_j + b_j$, $s_j' = a_j s_j$: $(a_j x_j + b_j - \mu_j')/s_j' = (x_j - \mu_j)/s_j$. (ii) Linearity of expectation gives mean 0; the variance is the average of all $|G|^2$ entries of

the correlation matrix, whose diagonal contributes $|G|^{-1}$ and whose off-diagonal contributes $(1-|G|^{-1})\bar{\rho}$, hence Var $= |G|^{-1}+(1-|G|^{-1})\bar{\rho} \in [\bar{\rho}, 1]$. (iii) A non-negative combination of non-decreasing functions is non-decreasing. □

Remark 1. *Sign orientation σ_j is essential: without it, averaging a risk-increasing and a risk-decreasing feature cancels the signal. An earlier implementation omitted both orientation and any restriction to the concept space, concatenating the three concepts with the* first $\min(d_s, d_t)$ *raw columns by position; positional columns have unrelated meanings across sectors, which breaks the premise of* (1)*. The corrected design uses $\mathcal{Z} = \mathbb{R}^3$ (and, optionally, a shared set of engineered universal features such as the robust amount z-score and a velocity proxy).*

C. Class imbalance and decision rule

Let $p(x) = \mathbb{P}(Y = 1 \mid x)$ under the deployment prior π.

a) Weighted loss.: With positive-class weight $\omega > 0$ the empirical risk is $\hat{R}_\omega(f) = \frac{1}{n}\sum_i [\omega y_i \ell(f(x_i), 1) + (1 - y_i)\ell(f(x_i), 0)]$ where ℓ is the logistic loss.

Proposition 3 (Weighting is a prior shift)**.** *The population minimiser of R_ω over all measurable f yields probabilities $q(x) = \sigma(f^\star(x))$ with*

$$\frac{q(x)}{1-q(x)} = \omega \, \frac{p(x)}{1-p(x)}. \tag{5}$$

Consequently $p(x) = q(x)/\big(q(x) + \omega(1-q(x))\big)$. The same relation holds, with $\omega = \frac{\pi'/(1-\pi')}{\pi/(1-\pi)}$, for training on data resampled by replication to prior π' (and approximately for SMOTE, whose synthetic points leave the class-conditional density only approximately unchanged [21], [22]).

Proof. Pointwise, the minimiser of $p\,\omega\,\ell(f,1) + (1-p)\,\ell(f,0)$ is the logistic-calibrated value for the normalised weights $\tilde{p} = \omega p/(\omega p + 1 - p)$, i.e. $\sigma(f^\star) = \tilde{p}$. Odds scale by ω. □

b) SMOTE.: For a minority point x_i with a k-nearest minority neighbour x_ν, SMOTE generates $x' = x_i + \gamma(x_\nu - x_i)$, $\gamma \sim \mathrm{U}[0,1]$ [21]. Synthetic points lie in the convex hull of minority neighbours; applied before data splitting or inside cross-validation *outside* the fold loop it places near-copies of validation points in training folds, inflating scores. We apply it inside each training fold.

Proposition 4 (Cost-sensitive Bayes threshold)**.** *With costs $c_{FP}, c_{FN} > 0$ for false positives and false negatives (zero for correct decisions), the decision minimising expected cost predicts fraud iff $p(x) \geq \tau^\star = c_{FP}/(c_{FP} + c_{FN})$. In terms of a model trained with weight ω, the equivalent threshold on q is $\tau_q = \omega\tau^\star/(\omega\tau^\star + 1 - \tau^\star)$.*

Proof. Predicting 1 costs $(1-p)c_{FP}$, predicting 0 costs $p\,c_{FN}$; the first is no larger iff $p \geq c_{FP}/(c_{FP} + c_{FN})$. Substituting Prop. 3 gives τ_q. □

Operationally, thresholds are chosen on a *validation* partition (never on test) either by maximising F_1/MCC or, preferably, by Prop. 4 with sector-specific costs.

D. Within-domain models

Logistic regression (L2-regularised). **Random forest** [23]: bagged trees, $\hat{p}(x) = \frac{1}{B}\sum_b T_b(x)$. **Gradient boosting** [3] minimises $\sum_i \ell(y_i, \hat{y}_i) + \sum_m \Omega(f_m)$ with $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2$; with second-order Taylor expansion each leaf weight is $w_j^\star = -G_j/(H_j + \lambda)$ and the split gain is

$$\tfrac{1}{2}\Big[\tfrac{G_L^2}{H_L+\lambda} + \tfrac{G_R^2}{H_R+\lambda} - \tfrac{(G_L+G_R)^2}{H_L+H_R+\lambda}\Big] - \gamma. \tag{6}$$

Isolation forest [4]: anomaly score $s(x,n) = 2^{-\mathbb{E}[h(x)]/c(n)}$ with $c(n) = 2H(n-1) - 2(n-1)/n$, H the harmonic number. **Autoencoder**: trained on legitimate training points only, with score $\|x - \psi(\varphi(x))\|_2^2$.

E. Domain adaptation

After alignment, source and target samples live in $\mathcal{Z}$. Let $G_f : \mathcal{Z} \to \mathbb{R}^m$ be a feature extractor and $G_y : \mathbb{R}^m \to [0,1]$ a classifier. All three methods minimise

$$\min_{G_f, G_y} \;\; \hat{R}_s^\omega(G_y \circ G_f) + \alpha\,\mathcal{L}_{\text{align}}(G_f) + \beta\,\hat{R}_{t,b}^{\omega_t}(G_y \circ G_f), \tag{7}$$

where $\hat{R}_s^\omega$ is the weighted source risk, $\hat{R}_{t,b}$ the risk on the b labelled target examples ($\beta = 0$ when $b = 0$), and $\mathcal{L}_{\text{align}}$ differs by method.

a) DANN.: Add a domain classifier $G_d : \mathbb{R}^m \to [0,1]$ with domain labels $d = 0$ (source), $d = 1$ (target):

$$\begin{aligned} &E(\theta_f, \theta_y, \theta_d) = \hat{R}_s^\omega \\ &-\lambda\Big[\tfrac{1}{n_s}\sum_i \mathcal{L}_d\big(G_d(G_f(x_i^s)), 0\big) + \tfrac{1}{n_t}\sum_j \mathcal{L}_d\big(G_d(G_f(x_j^t)), 1\big)\Big] \end{aligned} \tag{8}$$

and seek the saddle point $(\hat{\theta}_f, \hat{\theta}_y) = \arg\min E(\cdot, \cdot, \hat{\theta}_d)$, $\hat{\theta}_d = \arg\max E(\hat{\theta}_f, \hat{\theta}_y, \cdot)$ [7]. The adaptation weight follows $\lambda_p = \frac{2}{1+e^{-\gamma p}} - 1$, $p \in [0,1]$ training progress, $\gamma = 10$.

Proposition 5 (Gradient reversal realises the saddle point)**.** *Define a layer R_λ with $R_\lambda(z) = z$ and $\partial R_\lambda/\partial z = -\lambda I$. Minimising $\hat{R}_s^\omega + \mathcal{L}_d$ with the domain head fed through R_λ updates θ_d with $+\nabla_{\theta_d}\mathcal{L}_d$ (descent on $\mathcal{L}_d$) and θ_f with $\nabla_{\theta_f}\hat{R}_s^\omega - \lambda\nabla_{\theta_f}\mathcal{L}_d$, i.e. stochastic descent on E for (θ_f, θ_y) and ascent on E for θ_d (since $-\lambda\mathcal{L}_d$ enters E with negative sign, minimising $\mathcal{L}_d$ in θ_d maximises E).*

Proof. The chain rule through R_λ gives $\partial\mathcal{L}_d/\partial\theta_f = -\lambda\,(\partial\mathcal{L}_d/\partial z)(\partial z/\partial\theta_f)$ while $\partial\mathcal{L}_d/\partial\theta_d$ is unaffected. □

Remark 2. *The reversal coefficient must appear* once*: either in the layer (with unit weight on $\mathcal{L}_d$) or as an explicit loss weight with a plain identity layer, not both. An earlier implementation multiplied the domain loss by λ* and *reversed with $-\lambda$, so the extractor saw $-\lambda^2\nabla\mathcal{L}_d$ and the discriminator $\lambda\nabla\mathcal{L}_d$, silently changing the schedule; the corrected objective applies it once.*

By (1), driving a good domain discriminator to chance reduces the proxy $\mathcal{A}$-distance $d_{\mathcal{A}} = 2(1 - 2\epsilon_d)$, where ϵ_d is the discriminator's error, which we report as a diagnostic.

Proposition 6 (Label shift limits invariant representations [1])**.** *Let* $\mathrm{JS}(\cdot,\cdot)$ *denote the Jensen–Shannon divergence with* $\log_2$ *and* $d_{\mathrm{JS}} = \sqrt{\mathrm{JS}} \in [0,1]$*. For any representation* $Z = G_f(X)$ *and classifier, the sum of source and target errors satisfies*

$$\varepsilon_s + \varepsilon_t \;\geq\; \tfrac{1}{2}\big(d_{\mathrm{JS}}(\mathcal{D}_s^Y, \mathcal{D}_t^Y) - d_{\mathrm{JS}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z)\big)^2. \tag{9}$$

In particular, perfect marginal alignment ($d_{\mathrm{JS}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z) = 0$) *forces* $\lambda^\star$ *in* (1) *to be at least* $\frac{1}{2}\,\mathrm{JS}(\mathcal{D}_s^Y, \mathcal{D}_t^Y)$.

Applying this to the Bernoulli priors of Table I: Hence

TABLE II
LOWER BOUND ON $\varepsilon_s + \varepsilon_t$ UNDER PERFECT MARGINAL ALIGNMENT, $\frac{1}{2}\,\mathrm{JS}(\mathrm{Bern}(\pi_s), \mathrm{Bern}(\pi_t))$.

Pair	JS (bits)	d_{JS}	bound
CC↔MM	0.3310	0.575	0.166
CC↔HI	0.2295	0.479	0.115
MM↔HI	0.0127	0.113	0.006

marginal alignment alone is expected to be harmful for transfers involving credit cards, and benign between mobile money and insurance. This is a testable prediction (Section VII); the remedy is to estimate target priors and importance-weight the source (label-shift correction) or to rely on target labels (HEN).

b) Subdomain (class-conditional) MMD.: Let k be a Gaussian kernel $k_\sigma(u,v) = \exp(-\|u-v\|^2/2\sigma^2)$ with RKHS $\mathcal{H}_k$ and feature map ψ. The (biased) squared maximum mean discrepancy between samples $\{u_i\}_1^m$, $\{v_j\}_1^n$ is

$$\widehat{\mathrm{MMD}}^2 = \frac{1}{m^2}\sum_{i,i'} k(u_i,u_{i'}) + \frac{1}{n^2}\sum_{j,j'} k(v_j,v_{j'}) - \frac{2}{mn}\sum_{i,j} k(u_i,v_j). \tag{10}$$

Proposition 7 (MMD properties [10])**.** $\mathrm{MMD}^2(P,Q) = \|\mathbb{E}_P\psi(u) - \mathbb{E}_Q\psi(v)\|^2_{\mathcal{H}_k} = \sup_{\|f\|_{\mathcal{H}_k}\leq 1}\big(\mathbb{E}_P f - \mathbb{E}_Q f\big)^2$*. For a characteristic kernel (Gaussian on* $\mathbb{R}^d$*),* $\mathrm{MMD}(P,Q) = 0 \iff P = Q$*; the unbiased estimator excluding diagonal terms has expectation* MMD^2.

Global MMD matches marginals and therefore inherits Prop. 6. The *class-conditional* (subdomain) variant matches conditionals $p(z \mid y = c)$ [12]:

$$\mathcal{L}_{\mathrm{LMMD}} = \frac{1}{C}\sum_{c=1}^{C}\Big\| \sum_i w_{ic}^s\,\psi(G_f(x_i^s)) - \sum_j w_{jc}^t\,\psi(G_f(x_j^t))\Big\|^2_{\mathcal{H}_k}, \tag{11}$$

with $w_{ic}^s = y_{ic}/\sum_{i'} y_{i'c}$ (one-hot source labels) and $w_{jc}^t = \hat{y}_{jc}/\sum_{j'}\hat{y}_{j'c}$ using soft predictions; with only $C = 2$ classes and a rare positive class, the fraud weights are carried by few examples, so we use a multi-bandwidth kernel $\sum_u k_{\sigma_u}$, $\sigma_u \in \{0.5, 1, 2, 4\}$ and report per-class sample counts. Because the conditional distributions are matched, the bound of Prop. 6 does not apply to $\mathcal{L}_{\mathrm{LMMD}}$ to the same extent. (We note that the clustering-based "subdomain" described in an earlier README is *not* what the notebook implemented: it computed a single global kernel MMD on one hidden layer with a fixed bandwidth. The present formulation is class-conditional by definition.)

c) HEN (attention-gated pre-train/fine-tune).: A feature-attention gate $a(z) = \mathrm{softmax}\big(W_2\tanh(W_1 z + b_1) + b_2\big) \in \Delta^{d-1}$ re-weights inputs, $\tilde{z} = z \odot a(z)$, followed by an encoder g_ϕ and a linear head: $h(z) = \sigma(w^\top g_\phi(\tilde{z}) + b)$. Stage 1 minimises $\hat{R}_s^\omega$ ($\beta = 0$). Stage 2 fine-tunes on the b labelled target examples from the warm start θ_s with a smaller step and an L2-SP anchor [24]:

$$\hat{\theta}_t = \arg\min_\theta\; \hat{R}_{t,b}^{\omega_t}(\theta) + \tfrac{\eta}{2}\|\theta - \theta_s\|_2^2. \tag{12}$$

Remark 3. *Because the softmax sums to one, each gated coordinate is scaled by roughly* $1/d$*; batch normalisation after the first linear layer removes the resulting scale effect, but attention weights are then comparable only within, not across, models. Attention is a model-internal diagnostic, not a faithful explanation; explanations are obtained with SHAP (Section V-G). Since HEN uses target labels, it competes in the label-budget setting* $b > 0$ *against a* target-only *model trained on the same* b *labels; Section VI defines the transfer gain.*

F. Stacked ensemble

Base learners $f_1, \ldots, f_M$ (LR, RF, XGB) produce out-of-fold predictions $z_{im} = f_m^{(-\kappa(i))}(x_i)$ where $\kappa(i)$ is the fold of i; a logistic meta-learner is fit on $z_i = (z_{i1}, \ldots, z_{iM})$ [25]. Resampling must be inside the fold loop so that OOF predictions do not see synthetic neighbours of the held-out point.

G. Explainability

For a model f and feature set F ($|F| = M$), the Shapley attribution of feature j at x is

$$\phi_j(x) = \sum_{S\subseteq F\setminus\{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!}\big[f_{S\cup\{j\}}(x) - f_S(x)\big], \tag{13}$$

the unique allocation satisfying efficiency ($\sum_j \phi_j = f(x) - \mathbb{E}f$), symmetry, dummy and additivity [26]; TreeSHAP computes it exactly for tree ensembles in $O(TLD^2)$. LIME fits $g = \arg\min_{g\in\mathcal{G}} \mathcal{L}(f, g, \pi_x) + \Omega(g)$ locally [27]. For cross-sector comparison, importances are aggregated to the three concepts, $I_c = \sum_{j\in G_{kc}} \overline{|\phi_j|}$ normalised to sum to one per sector, and compared by Kendall's τ with a permutation test; raw feature importances are *not* comparable across sectors because feature sets differ.

VI. EXPERIMENTAL PROTOCOL

a) Splits.: CC and MM: chronological 70/15/15 by transaction time (Sparkov's published train/test boundary is respected; it was previously merged and re-split at random). HI: grouped stratified split by patient identifier.

b) Regimes.: (R1) within-domain; (R2) zero-shot transfer ($b = 0$); (R3) few-shot transfer with $b \in \{10, 50, 100, 500\}$ labels, drawn with 20 random seeds; (R4) *target-only* baseline using the same b labels. The *transfer gain* of method m at budget b is

$$\Delta_m(b) = M\big(\hat{h}_m^{(b)}\big) - M\big(\hat{h}_{\text{target-only}}^{(b)}\big), \tag{14}$$

for a metric M; positive Δ indicates real transfer rather than label supervision alone.

c) Model selection and thresholds.: No target-test labels, and for $b = 0$ no target labels at all, are used for early stopping, checkpoint choice or thresholds. Hyper-parameters are selected by (a) source validation, (b) importance-weighted validation [16], or (c) for $b > 0$, cross-validation within the b labelled target examples. Thresholds follow Prop. 4.

d) Metrics.: With confusion counts TP, FP, TN, FN (MCC [28]; precision–recall analysis [29]):

$$F_1 = \frac{2TP}{2TP + FP + FN}, \tag{15}$$

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \tag{16}$$

$$\mathrm{AUC} = \mathbb{P}\big(s(X^+) > s(X^-)\big) + \tfrac{1}{2}\mathbb{P}\big(s(X^+) = s(X^-)\big), \tag{17}$$

$$\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n. \tag{18}$$

Average precision (the step-wise sum) is reported instead of the trapezoidal area $\mathrm{auc}(R, P)$, which interpolates linearly between operating points and is optimistic. Because π differs by two orders of magnitude between sectors, metrics are reported in *all* sectors (not a different metric per sector), alongside the expected cost of Prop. 4 and precision at a fixed alert budget.

e) Statistics.: Over K seeds/folds the paired difference $d_i = M_i^A - M_i^B$ has estimate $\bar{d}$ and sample variance $\hat{\sigma}_d^2$. Repeated resampling creates correlated estimates, so we use the corrected resampled t-statistic [15]

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{K} + \frac{n_{\mathrm{te}}}{n_{\mathrm{tr}}}\right)\hat{\sigma}_d^2}}, \tag{19}$$

with Wilcoxon signed-rank as a non-parametric check, DeLong's test for correlated AUCs [30], bootstrap percentile intervals ($B = 1000$), and Holm's step-down correction [31]: sort $p_{(1)} \le \cdots \le p_{(m)}$ and reject $H_{(i)}$ while $p_{(i)} \le \alpha/(m-i+1)$. Comparisons are never pooled across heterogeneous transfer scenarios (a paired test over the six scenario scores treats unlike units as exchangeable); across-scenario summaries use the Friedman test with Nemenyi post-hoc [32].

f) Ablations and baselines.: (A1) remove alignment (raw positional features, shown for contrast); (A2) alignment without orientation σ_j; (A3) each adaptation loss removed; (A4) global vs. class-conditional MMD; (A5) label-shift-corrected source weights; (A6) CORAL [11]; (A7) gradient boosting with a domain indicator; (A8) tabular foundation models [33] and FT-Transformer [34] as strong few-label baselines.

VII. Results

All numbers in this section are placeholders; the prior run is not reported because its pipeline fits target encodings, outlier caps, scalers and feature selection on the full dataset, computes a label-derived per-DRG rate before splitting, selects checkpoints and thresholds on target labels, and retains quasi-identifiers in the health-insurance features. Under those conditions reported health-insurance scores near $F_1 \approx 0.996$ are not interpretable (Prop. 1).

TABLE III
Within-domain results (R1), test set, mean ± 95% bootstrap CI over seeds.

Sector	Model	AP	MCC	Cost
CC	LR / RF / XGB / Stack	**TBD**	**TBD**	**TBD**
MM	LR / RF / XGB / Stack	**TBD**	**TBD**	**TBD**
HI	LR / RF / XGB / Stack	**TBD**	**TBD**	**TBD**

TABLE IV
Transfer gain $\Delta_m(b)$ in AP over the target-only baseline; rows are directed pairs, columns label budgets.

Pair	Method	b=0	10	100	500
CC→MM	DANN	**TBD**	**TBD**	**TBD**	**TBD**
	LMMD	**TBD**	**TBD**	**TBD**	**TBD**
	HEN	–	**TBD**	**TBD**	**TBD**
… (all six directed pairs)					

Predictions to be confirmed or refuted. (P1) DANN and global MMD underperform the target-only baseline for CC→MM and CC→HI, where Table II predicts an irreducible error floor, and are neutral for MM↔HI. (P2) Class-conditional LMMD and label-shift-corrected weights narrow this gap. (P3) Gains of HEN vanish once compared with the target-only baseline at equal b unless the source representation is genuinely informative. (P4) Concept-level SHAP rankings agree more strongly between MM and HI than with CC.

VIII. Discussion and Threats to Validity

Synthetic data. Two sectors are simulated; conclusions about relative method ranking may not transfer to production traffic, and simulators encode the generator's own fraud heuristics (e.g. balance-consistency features are close to label-defining in MoMTSim) [35]. **Label validity (HI).** Rule labels measure tariff non-compliance, not adjudicated fraud, and are circular with billing features (Prop. 1); the missing-drug-charge artefact in r_5 (Section IV) must be removed or reported separately. **Small n.** Health insurance has $n = 2655$; confidence intervals are wide and few-shot results are high-variance. **Concept map.** Group membership G_{kc} and orientations σ_j are expert-specified; sensitivity to these choices is an explicit ablation (A2). **Fairness and harm.** Flagging rates should be audited across region and facility type; thresholds should reflect investigation capacity.

IX. Conclusion

We gave a complete mathematical specification of a heterogeneous cross-sector fraud-transfer pipeline: an affine-invariant semantic alignment, a unified view of adversarial, kernel and fine-tuning adaptation, prior-aware decision theory, and an evaluation protocol free of the leakage channels that most threaten such studies. The label-shift analysis predicts concrete failure modes of invariant-representation methods for Ghana's sector pairs, which the corrected experiments will test.

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